Banknote Authenticity Detection using Computer Vision

1. Problem Statement

Detecting counterfeit banknotes is a relevant real-world problem in financial security. Manual inspection is not always reliable, especially for non-experts.

This project aims to automate the classification of banknotes as genuine or fake using Computer Vision techniques.

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2. Methodology

The project explores two different approaches:

2.1 Classical Computer Vision Approach

Preprocessing

- Grayscale conversion
- Noise reduction (Gaussian blur)
- Edge detection
- Contour detection
- Perspective transformation

Feature Extraction

- HOG → shape features
- LBP → texture features
- Color histograms
- Statistical descriptors

Classification

- Support Vector Machine (SVM)

2.2 Deep Learning Approach

A Convolutional Neural Network (CNN) is implemented using **MobileNetV2**.

- Pre-trained on ImageNet
- Feature extractor layers are frozen
- Final layer is replaced for binary classification

This approach allows automatic feature learning instead of manual feature engineering.

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3. Experimental Results

Both approaches are evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Observation: The classical SVM model performs better than the CNN model.

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4. Failure Analysis

Several limitations affect the system:

- Small dataset size
- Artificial fake images do not reflect real counterfeits
- CNN requires larger datasets to perform well
- Preprocessing may fail in difficult lighting conditions

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5. Ethical Considerations

- Dataset bias may affect results
- Incorrect classification may lead to wrong decisions
- The system should not be used in critical real-world scenarios without further validation

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6. Conclusion

This project demonstrates two different approaches to image classification.

The results show that:

- Classical methods can perform well on small datasets
- Deep learning requires more data but offers better scalability

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